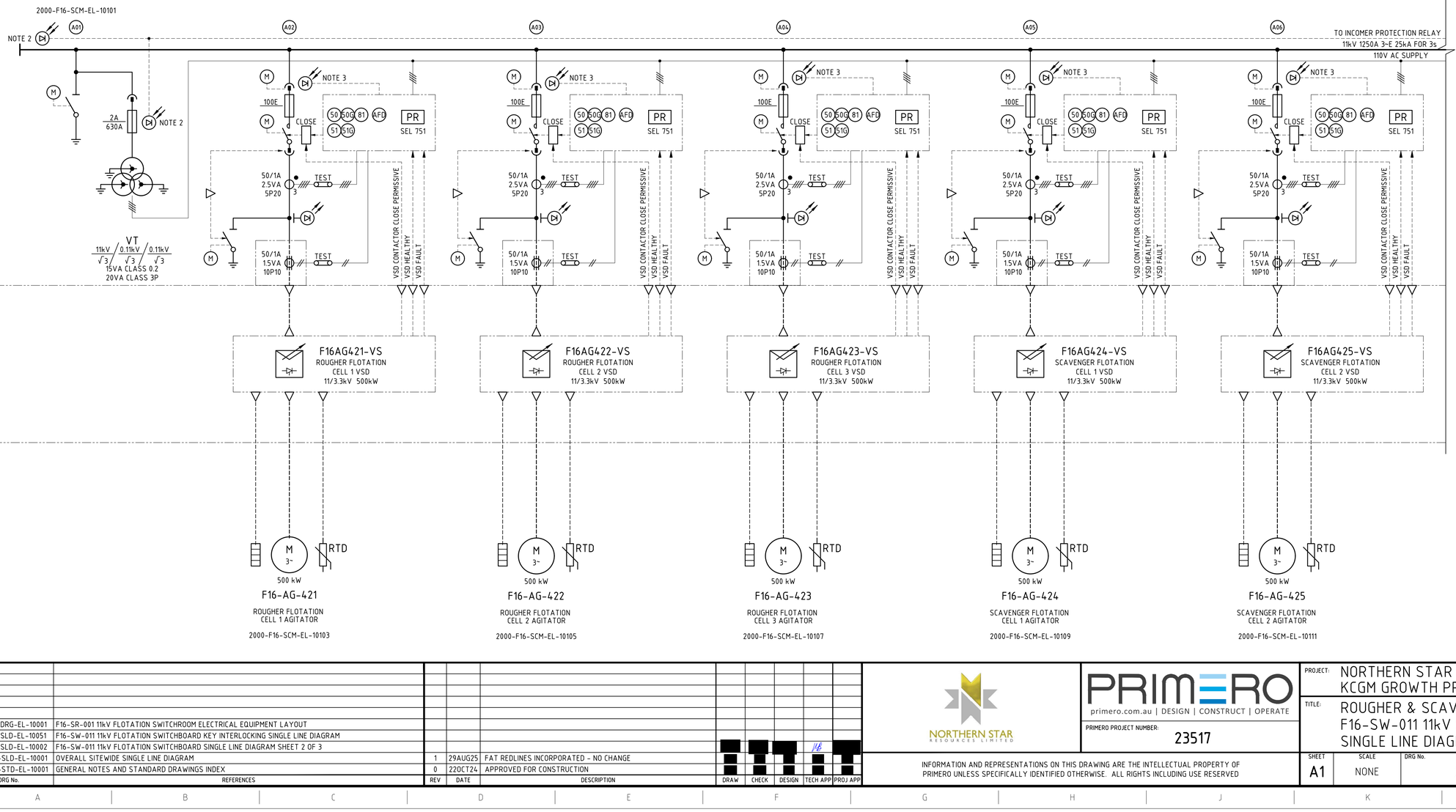
F16-SR-001
11kV FLOTATION SWITCHROOM

F16-SW-011
FLOTATION 11kV SWITCHBOARD



NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 23/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

DRG No.	REFERENCES
2000-F16-DRG-EL-10001	F16-SR-001 11kV FLOTATION SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F16-SLD-EL-10051	F16-SW-011 11kV FLOTATION SWITCHBOARD KEY INTERLOCKING SINGLE LINE DIAGRAM
2000-F16-SLD-EL-10002	F16-SW-011 11kV FLOTATION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 2 OF 3
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	29AUG25	FAT REDLINES INCORPORATED - NO CHANGE	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
0	22OCT24	APPROVED FOR CONSTRUCTION	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER & SCAVENGER FLOTATION
F16-SW-011 11kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SLD-EL-10001	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10001.DWG 09SEP25 6:50AM HOTSEAT04